Chengwen QU

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Open to relocation

Canadian citizen

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Tech blog | Github

SUMMARY

Software engineer with a unique background spanning engineering and modern software development. Experienced in object-oriented design, cloud native solution engineering, database schema design, I built data-intensive enterprise system, empowering hundreds of organizations. My work has consistently translated into real world impacts, with a proven track record of delivering solutions from proof of concepts through to production.

EDUCATION

Memorial University of Newfoundland (2013 - 2018)

Bachelor of Engineering, Ocean and Naval Architectural Engineering

Georgia Institute of Technology (2027 Dec, part time OMSCS)

Master of Computer Science, specialization in ML

Additional Info

Awards/Training: AWS Developing Serverless Solutions, AWS DevOps Engineering, Nvidia Fundamentals of Deep Learning, Nvidia Building AI-Based Cybersecurity Pipelines

Skills: C, Java, Python, Go, SQL, Object-Oriented Design, Data Storage Design, Git, Bash Scripting, Groovy, Gradle, Apache Kafka, Postgres, Redis, ECS Fargate, AWS Step Function, DynamoDB, Terraform, Kubernetes

WORK EXPERIENCE

Trend AI (Trend Micro Canada) (2021 – present)

Senior Software Developer – Tech Lead (2026.7-present)

- Lead development of new features in collaboration with product management team
- Lead performance engineering initiatives for compliance management service in Trend Vision One platform

Senior Software Developer (2023.2-present)

- Performed systems analysis, including requirement clarification, solution design, and impact assessment for system changes
- Developed an agentic LLM-based compliance agent that generates actionable, dashboard-level overview of customer's security posture
- Designed and architected validator service in the new alerting microservice
- Identified and eliminated processing bottlenecks in high volume workloads
- Optimized DB performance through indexing, partitioning, query refactoring and caching strategies
- Troubleshooted issues and provide solutions related to forwarding of SIEM events for customers
- Improved CI/CD pipelines, testing, and deployment processes to ensure high-quality releases
- Mentored junior developers and co-op student

Backend Software Developer (2021.11- 2023.2)

- Developed and maintained large scale data-pipeline services enabling near real-time data integration across multiple databases and storage systems using Kafka
- Improved deployment automation using groovy
- Wrote non-application code, such as infrastructure as code, pipeline as code, unit tests, metrics/alarms, monitoring etc.

TechnipFMC Canada (2018 – 2020)

Project Engineer

- Performed engineering work related to design, procurement, manufacturing and installation of offshore oil production structures

Notable Personal Projects

Stock analysis and trading website (2020)

- A virtual stock trading website where users can browse stocks and their real-time price, buy and sell stocks

Android App (2021)

- pick dates and browse 10,000+ NASA images and its feature, vote Favorites

Software defined network (2023)

- Router design and algorithm, BGP hijacking

Multi-threaded http library – System programming (2024)

- Multi-threaded http-based server client library written in C (see blog)

Data Visualization App

- Visualizing Canada federal election results